

Jaiaditya Asudani

Github: github.com/Jai2379
Portfolio: [Jai2379.github.io](https://jai2379.github.io)

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EDUCATION

- University of Waterloo** Waterloo, ON
 - Candidate for BAsC in Nanotechnology Engineering* *September 2025 – Present*
Expected Graduation: May 2030
 - Relevant Coursework: Materials Science & Nanomaterials, Electricity, Magnetism & Circuits, Computational Methods*

SKILLS SUMMARY

- Design & Modeling:** SolidWorks, AutoCAD, CATIA, Fusion 360, OnShape, GD&T, FEA
- Computational Tools:** Python (NumPy, Pandas), MATLAB, C/C++, Git/GitHub
- Embedded Systems:** Micro-controllers (Arduino), Embedded Sensors, Circuit Design
- Fabrication & Prototyping:** CNC, Machining (Mill, Lathe), Carbon Fiber, Rapid Prototyping

EXPERIENCE

- UW NanoRobotics Group (Design Team) — University of Waterloo** Waterloo, Ontario
 - Robotics Sub-team Member* *January 2026 - Present*
 - Optimized **CNC deposition** of water-based nanoparticle inks to improve circuit trace precision for additive **PCB manufacturing**.
 - Debugged motion control firmware (**C/C++**) to refine multi-axis displacement accuracy and ensure consistent ink flow.
 - Validated sensor integration and **microcontroller** communication to automate real-time feedback during the micro-fabrication process.
- Waterloop (Design Team) — University of Waterloo** Waterloo, Ontario
 - Mechanical Engineering Member* *October 2025 - January 2026*
 - Integrated **20+** mechanical components into a full-scale hyperpod assembly using **SolidWorks** and **Onshape**, resolving fit, alignment, and subsystem interface constraints.
 - Designed and iterated **3 mechanical components** across multiple revision cycles, applying **DFM** principles for manual machining (mill, lathe) and basic CNC fabrication.
 - Conducted carbon fiber composite materials research, evaluating strength-to-weight tradeoffs to support lightweight structural design decisions. **7% reduction** in weight.
- Electrium Mobility (Design Team) — University of Waterloo** Waterloo, Ontario
 - Mechanical Engineering Member* *October 2025 - January 2026*
 - Modeled and assembled structural and mounting components for a **full-scale** electric bike in SolidWorks, contributing to system-level mechanical integration.
 - Designed and integrated **3+ human-machine interface components** (throttle housing, handlebar elements, display mounts) with emphasis on ergonomics and durability.
 - Applied **DFM principles**, accounting for **tolerance stack-up**, fastener access, and assembly sequencing to improve prototype reliability.
- Mathematics Tutor** Brampton, Ontario
 - Private/Independent* *February 2025 - May 2025*
 - Delivered weekly 1:1 tutoring sessions over **4+ months**, resulting in a **13%** average improvement in students' mathematics grade.
 - Developed custom lesson plans to strengthen problem-solving and logical reasoning in advanced mathematics.
 - Independently managed lesson planning, scheduling, and progress tracking across a multi-month timeframe.

PROJECTS

- TENG Device (Tribo-Electric Nano Generator):** Designed and fabricated a triboelectric nanogenerator (TENG) during a 2-day rapid prototyping challenge, achieving 22 V electrical output as a proof-of-concept. — **SolidWorks, Engineering-Design**
- 6-DOF Robotic Arm (Prototype):** Engineered a 6-degree-of-freedom robotic arm in SolidWorks, developing a fully articulated mechanical assembly with defined motion constraints. Designed for modularity and scalability, enabling future Arduino-based actuation and control integration. — **SolidWorks, 3D Printing**
- Adaptive Presence-Based Lighting Control System (WIP):** Developing a smart lighting system using an Arduino Uno and PIR motion sensor to dynamically regulate lighting based on user presence. Implementing embedded C++ logic and digital I/O control to improve energy efficiency during inactivity. — **Arduino, Motion-Detection, C++, Circuits**
- Interactive Data Analysis Tool:** Developed an interactive histogram and implemented dynamic binning, distribution fitting, and user-controlled parameters to support exploratory data analysis and statistical insight. — **Python, NumPy/SciPy, Streamlit**